Data Dictionary

Every table and column of the database of Refrigerator Factory (Version 2.0), as built. The names, types, keys and defaults are read from `backend/prisma/schema.prisma`; the descriptions say what each one is for. The relations between the tables are in the ER Diagram and the keys and delete rules in the DB Relations.

How to read it

Item	Meaning
Table, column	The names in the database, as the Prisma model gives them (a column is written in camel case, for example `passwordHash`)
Type	The MySQL type: `varchar(191)` text, `text` long text, `int` whole number, `decimal(14,3)` quantity with 3 decimals, `date`, `datetime(3)`, `boolean`
Null	`yes` when the column may be empty
Key	`PK` primary key, `FK` foreign key (the table it points to is in the description), `UK` unique
Default	The value used when none is given: `auto` is given by the database, `now` is the current time
Money	Whole yen in an `int`; no tax, no decimals
Quantity	`decimal(14,3)` for materials; whole `int` for products

Value lists

These columns hold text, but only the values listed here.

Column	Values	Meaning
`User.role`	`Admin`, `Worker`	Admin does everything. A Worker works by department.
`User.department`	`Sales`, `Factory`, `Accounting`, `Administration`	A Worker must be `Sales` or `Factory`; the department decides the access.
`QtReference.status`	`Draft`, `Confirmed`	Draft can be changed and deleted; Confirmed is locked and has one production order.
`OrderProcess.status`	`Not Started`, `In Progress`, `Completed`	Not Started means quantity 0, Completed means the full quantity of the line.
`Invoice.status`	`Unpaid`, `Paid`	Paid after one full payment, which also issues the receipt.
`PurchaseOrder.status`	`Draft`, `Sent`, `Received`	Moved by hand, one step forward. Received adds the stock once.
`StockChange.kind`	`Add`, `Deduct`, `Adjust`	Add and Deduct move a lot by a quantity; Adjust is a signed correction and needs a reason.
`Attachment.kind`	`PO_FILE`, `QUOTATION`, `EVIDENCE`, `OTHER`	What the file is. A new `PO_FILE` replaces the old one.
`Attachment.ownerType`	`QtReference`, `ProductionOrder`, `PurchaseOrder`	The table the file belongs to (with `ownerId`).

Column	Values	Meaning
`PurchasePayment.method`	`Cash`, `Bank transfer`, `Cheque`, `Credit card`, `Other`	How the purchase order was paid.

Tables

23 tables in 5 groups.

Group	Tables
Users and files	User, Attachment
Master data	Customer, Product, Material, ProductBom, MaterialLot, Supplier, SupplierMaterial
Sale	QtReference, QtLine, ProductionOrder, OrderLine, OrderProcess, ProgressEntry
Accounting	Invoice, InvoiceLine, Payment, Receipt
Purchase and stock	PurchaseOrder, PurchaseOrderLine, PurchasePayment, StockChange

Users and files

User

A person who signs in: Admin, Sale Worker or Factory Worker.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.
`name`	varchar(191)	no			Display name.
`department`	varchar(191)	no		Administration	Department of the user; decides the access of a Worker.
`email`	varchar(191)	no	UK		Sign-in name, stored in lower case. Unique.
`passwordHash`	varchar(191)	no			Hash of the password (bcrypt). The password itself is never stored.
`role`	varchar(191)	no		Worker	`Admin` or `Worker`.
`tokenVersion`	int	no		0	Grows with every new password; a token with an older number is refused, so old sessions end.
`lastLoginAt`	datetime(3)	yes			Time of the last sign-in.
`createdAt`	datetime(3)	no		now	When the row was created.
`updatedAt`	datetime(3)	no		auto	When the row was last changed.

Attachment

A file kept with a QT, a production order or a purchase order.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.
`ownerType`	varchar(191)	no			Table of the owner. There is no database key to it; the service checks it.

Column	Type	Null	Key	Default	Description
`ownerId`	int	no			Id of the owner row.
`kind`	varchar(191)	no			What the file is.
`fileName`	varchar(191)	no			Original name of the file.
`fileType`	varchar(191)	no			Media type: PDF, PNG or JPG.
`size`	int	no			Size in bytes (up to 10 MB).
`filePath`	varchar(191)	no			Name of the stored file in the upload folder.
`uploadedBy`	int	no	FK		The user who uploaded it. Points to `User`.
`at`	datetime(3)	no		now	When the row was written.

- No foreign key from `ownerType` + `ownerId`; the service checks the owner.

Master data

Customer

A company that buys from the factory.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.
`name`	varchar(191)	no			Company name.
`address`	text	no			Postal address, printed on the QT, invoice and receipt.
`email`	varchar(191)	no			Contact e-mail (may be empty).
`telephone`	varchar(191)	no			Contact telephone (may be empty).
`createdAt`	datetime(3)	no		now	When the row was created.
`updatedAt`	datetime(3)	no		auto	When the row was last changed.

Product

A product the factory makes and sells.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.
`code`	varchar(191)	no	UK		Product code. Unique.
`name`	varchar(191)	no			Product name.
`price`	int	no		0	Default unit price in yen; copied to a QT line and then editable there.
`createdAt`	datetime(3)	no		now	When the row was created.
`updatedAt`	datetime(3)	no		auto	When the row was last changed.

Material

A raw material or part kept in stock.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.
`code`	varchar(191)	no	UK		Material code. Unique.
`name`	varchar(191)	no			Material name.
`unit`	varchar(191)	no		pcs	Unit of measure (pcs, kg, ...).
`minLevel`	decimal(14,3)	yes			Minimum stock level. Empty means no level, so the material is never Low or Out.
`createdAt`	datetime(3)	no		now	When the row was created.
`updatedAt`	datetime(3)	no		auto	When the row was last changed.

ProductBom

Bill of materials: how much of a material one product needs. One row per product and material.

Column	Type	Null	Key	Default	Description
`productId`	int	no	PK, FK		The product. The row goes with the product when it is deleted. Points to `Product`.
`materialId`	int	no	PK, FK		The material. Points to `Material`.
`quantity`	decimal(14,3)	no			Quantity of the material for one product.

MaterialLot

A lot of a material in stock. The total of the lots is the stock of the material.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.
`materialId`	int	no	FK		The material. The lot goes with the material when it is deleted. Points to `Material`.
`lotNumber`	varchar(191)	no			Lot number, unique inside the material.
`quantity`	decimal(14,3)	no		0	Quantity left in the lot. Production takes from the oldest lot first.
`createdAt`	datetime(3)	no		now	When the row was created.

- Unique together: `materialId` + `lotNumber`.

Supplier

A company the factory buys materials from.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.
`code`	varchar(191)	no	UK		Supplier code. Unique.
`name`	varchar(191)	no			Supplier name.
`contact`	varchar(191)	no			Contact person or telephone (may be empty).

Column	Type	Null	Key	Default	Description
address	text	no			Postal address, printed on the purchase order.
active	boolean	no		true	False hides the supplier from new purchase orders; old ones stay.
createdAt	datetime(3)	no		now	When the row was created.
updatedAt	datetime(3)	no		auto	When the row was last changed.

SupplierMaterial

A material a supplier sells, with its price. One row per supplier and material.

Column	Type	Null	Key	Default	Description
supplierId	int	no	PK, FK		The supplier. The row goes with the supplier when it is deleted. Points to Supplier.
materialId	int	no	PK, FK		The material. Points to Material.
unitPrice	int	no			Initial unit price in yen; copied to a purchase order line, where it can be changed.

Sale

QtReference

A quotation (QT). Confirming it makes one production order.

Column	Type	Null	Key	Default	Description
id	int	no	PK	auto	Primary key, given by the database.
qtNumber	varchar(191)	no	UK		QT number. Unique.
qtDate	date	no			Date of the QT.
customerId	int	no	FK		The customer. Points to Customer.
notes	text	no			Free notes.
customerPoNumber	varchar(191)	no			Number of the customer's purchase order (may be empty).
customerPoDate	date	yes			Date of the customer's purchase order.
status	varchar(191)	no		Draft	Draft or Confirmed.
productionOrderId	int	yes	FK, UK		The production order made by Confirm. Unique, so a QT makes only one order. Points to ProductionOrder.
createdAt	datetime(3)	no		now	When the row was created.
updatedAt	datetime(3)	no		auto	When the row was last changed.

QtLine

A product line of a QT. One row per QT and product.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.
`qtId`	int	no	FK		The QT. The line goes with the QT when it is deleted. Points to `QtReference`.
`productId`	int	no	FK		The product. Points to `Product`.
`quantity`	int	no			Quantity quoted.
`unitPrice`	int	no			Unit price in yen. The amount (quantity x unit price) is computed, not stored.

- Unique together: `qtId` + `productId`.

ProductionOrder

The factory order made by Confirm QT. Its status (Draft, In Progress, Completed) is computed from its processes.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.
`orderNumber`	varchar(191)	no	UK		Order number (ORD-yyyy-0001), given by the server. Unique.
`qtId`	int	no	FK		The QT it was made from. Points to `QtReference`.
`customerId`	int	no	FK		The customer, copied from the QT. Points to `Customer`.
`orderDate`	date	no			Date of the order.
`notes`	text	no			Free notes.
`startedAt`	datetime(3)	yes			When production started. Empty before the start; after it the products and processes are locked.
`createdAt`	datetime(3)	no		now	When the row was created.
`updatedAt`	datetime(3)	no		auto	When the row was last changed.

OrderLine

A product line of a production order, copied from the QT line.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.
`orderId`	int	no	FK		The order. The line goes with the order when it is deleted. Points to `ProductionOrder`.
`productId`	int	no	FK		The product. Points to `Product`.
`quantity`	int	no			Quantity to make.
`unitPrice`	int	no			Agreed unit price in yen.

OrderProcess

One manual step of making a line (cutting, assembly, testing, ...), in order.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.
`lineId`	int	no	FK		The order line. The process goes with the line when it is deleted. Points to `OrderLine`.
`sequence`	int	no			Position of the step, from 1.
`name`	varchar(191)	no			Name of the step.
`instructions`	text	no			How to do the step (may be empty).
`status`	varchar(191)	no		Not Started	`Not Started`, `In Progress` or `Completed`.
`completedQuantity`	int	no		0	Quantity finished so far, from 0 up to the quantity of the line.

ProgressEntry

History of progress: every time a process is updated. Never changed or deleted.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.
`processId`	int	no	FK		The process. Points to `OrderProcess`.
`status`	varchar(191)	no			Status set by this entry.
`completedQuantity`	int	no			Quantity finished at that time.
`workDate`	date	no			Date of the work.
`note`	text	no			Note (needed when a Completed process is reopened).
`userId`	int	no	FK		Who recorded it. Points to `User`.
`at`	datetime(3)	no		now	When the row was written.

Accounting

Invoice

An invoice for a Completed order: a copy of the order as it was. One per order.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.
`invoiceNumber`	varchar(191)	no	UK		Invoice number (INV-yyyy-0001). Unique.
`invoiceDate`	date	no			Date of the invoice.
`orderId`	int	no	FK, UK		The order. Unique, so an order is billed once. Points to `ProductionOrder`.
`customerName`	varchar(191)	no			Customer name, copied when issued.
`customerAddress`	text	no			Customer address, copied when issued.
`total`	int	no			Total in yen.

Column	Type	Null	Key	Default	Description
`status`	varchar(191)	no		Unpaid	`Unpaid` or `Paid`.
`createdAt`	datetime(3)	no		now	When the row was created.

InvoiceLine

A line of an invoice, with the product data copied when it was issued.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.
`invoiceId`	int	no	FK		The invoice. The line goes with the invoice. Points to `Invoice`.
`productCode`	varchar(191)	no			Product code, copied.
`productName`	varchar(191)	no			Product name, copied.
`quantity`	int	no			Quantity billed.
`unitPrice`	int	no			Unit price in yen.
`amount`	int	no			Quantity x unit price.

Payment

The one full payment of an invoice.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.
`invoiceId`	int	no	FK, UK		The invoice. Unique: one payment per invoice. Points to `Invoice`.
`paymentDate`	date	no			Date of the payment.
`amount`	int	no			Amount in yen. Must equal the total of the invoice.

Receipt

The receipt issued when an invoice is paid.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.
`receiptNumber`	varchar(191)	no	UK		Receipt number (RC-yyyy-0001). Unique.
`invoiceId`	int	no	FK, UK		The invoice. Unique: one receipt per invoice. Points to `Invoice`.
`receiptDate`	date	no			Date of the receipt.
`customerName`	varchar(191)	no			Customer name, copied.
`customerAddress`	text	no			Customer address, copied.
`amount`	int	no			Amount received in yen.

Purchase and stock

PurchaseOrder

A purchase order (PO) to a supplier.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.
`poNumber`	varchar(191)	no	UK		PO number (PO-yyyy-0001). Unique.
`supplierId`	int	no	FK		The supplier. Points to `Supplier`.
`department`	varchar(191)	no			Department the purchase is for (a plain list, not a permission).
`requiredDate`	date	no			Date the goods are needed.
`notes`	text	no			Free notes.
`status`	varchar(191)	no		Draft	`Draft`, `Sent` or `Received`.
`total`	int	no		0	Sum of the line amounts, in yen.
`receivedAt`	datetime(3)	yes			When the status became Received; the stock was added then, once.
`createdBy`	int	no	FK		The user who made the PO. Points to `User`.
`createdAt`	datetime(3)	no		now	When the row was created.
`updatedAt`	datetime(3)	no		auto	When the row was last changed.

PurchaseOrderLine

A material on a purchase order. Lines change only while the PO is Draft.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.
`poId`	int	no	FK		The PO. The line goes with the PO when it is deleted. Points to `PurchaseOrder`.
`materialId`	int	no	FK		The material (one the supplier sells). Points to `Material`.
`quantity`	decimal(14,3)	no			Quantity ordered.
`unit`	varchar(191)	no			Unit of the material, copied.
`unitPrice`	int	no			Unit price in yen; starts from the supplier's price.
`amount`	int	no			Quantity x unit price.
`lotNumber`	varchar(191)	yes			Lot typed for this line when the PO is received; the received quantity goes into it.

PurchasePayment

The one payment of a received purchase order. It does not go through the Accounting tables.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.

Column	Type	Null	Key	Default	Description
`poId`	int	no	FK, UK		The PO. Unique: one payment per PO. Points to `PurchaseOrder`.
`paymentDate`	date	no			Date of the payment.
`amount`	int	no			Amount in yen. Must equal the total of the PO.
`method`	varchar(191)	no			How it was paid.
`referenceNo`	varchar(191)	no			Reference number of the payment (may be empty).
`note`	text	no			Free note.

StockChange

History of every change of a lot: by hand, by the start of an order, or by receiving a PO. Never changed or deleted.

Column	Type	Null	Key	Default	Description
`id`	int	no	PK	auto	Primary key, given by the database.
`materialId`	int	no	FK		The material. Points to `Material`.
`lotNumber`	varchar(191)	no			The lot that changed.
`kind`	varchar(191)	no			`Add`, `Deduct` or `Adjust`.
`change`	decimal(14,3)	no			Signed quantity added to the lot (negative for a deduction).
`balance`	decimal(14,3)	no			Quantity of the lot after the change.
`changeDate`	date	no			Date of the change.
`note`	text	no			Reason (needed for Adjust).
`initial`	boolean	no		false	True for the first stock of a material; a material with a later change cannot be deleted.
`orderId`	int	yes	FK		The production order whose start took the stock (if any). Points to `ProductionOrder`.
`purchaseOrderId`	int	yes	FK		The PO whose receiving added the stock (if any). Points to `PurchaseOrder`.
`userId`	int	no	FK		Who made the change. Points to `User`.
`at`	datetime(3)	no		now	When the row was written.